Week-03 HandsOn Solution

❖ WebApi

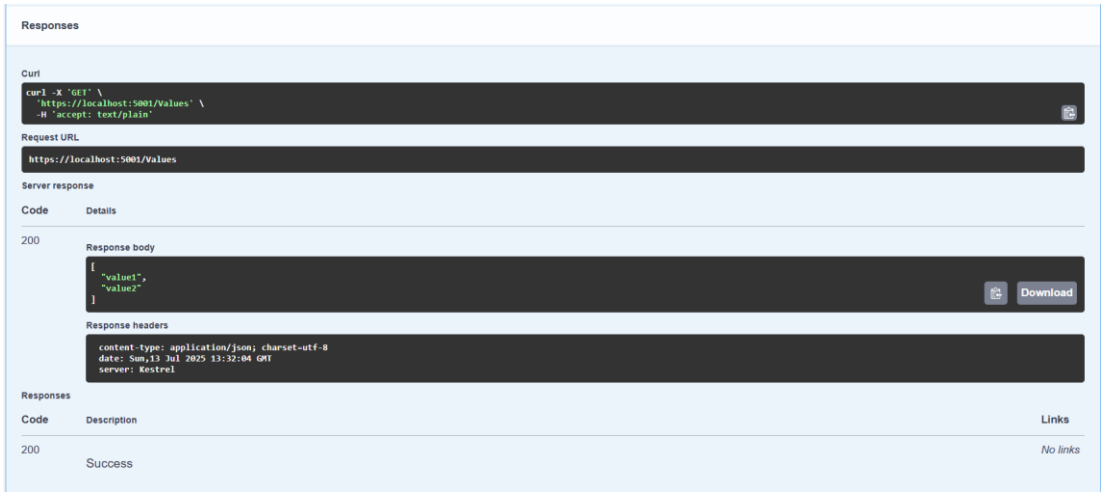
Exercise-1: Create a Web API with GET & POST

- **Code:**

```
using Microsoft.AspNetCore.Mvc;
using System.Collections.Generic;

namespace WebApiDemo.Controllers
{
  [ApiController]
  [Route("[controller]")]
  public class ValuesController : ControllerBase
  {
    private static List<string> values = new() { "value1", "value2" };

    [HttpGet]
    public IEnumerable<string> Get() => values;

    [HttpPost]
    public IActionResult Post([FromBody] string value)
    {
      values.Add(value);
      return Ok(values);
    }
  }
}
```

- **Output:**

//GET/Values



//POST/Values

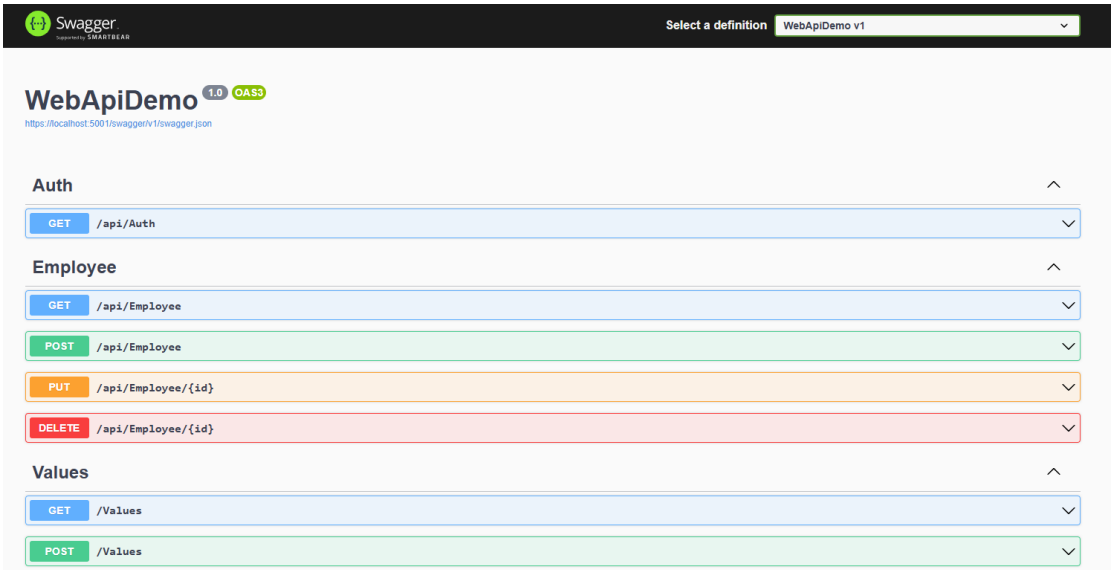
Exercise-2: Enable Swagger in Web API

- **Code**:

```
using Microsoft.AspNetCore.Authentication.JwtBearer;
using Microsoft.IdentityModel.Tokens;
using Microsoft.OpenApi.Models;
using System.Text;
using WebApiDemo.Filters;

var builder = WebApplication.CreateBuilder(args);

builder.Services.AddControllers();
builder.Services.AddEndpointsApiExplorer();

builder.Services.AddSwaggerGen(c =>
{
        c.SwaggerDoc("v1", new OpenApiInfo { Title = "WebApiDemo", Version = "v1" });

   c.AddSecurityDefinition("Bearer", new OpenApiSecurityScheme
   {
      Description = @"JWT Authorization header using the Bearer scheme.
                  Enter 'Bearer' followed by your token.
                  Example: Bearer eyJhbGciOiJIUzI1...",
      Name = "Authorization",
      In = ParameterLocation.Header,
      Type = SecuritySchemeType.ApiKey,
      Scheme = "Bearer"
        });

   c.AddSecurityRequirement(new OpenApiSecurityRequirement
   {
      {
         new OpenApiSecurityScheme
         {
            Reference = new OpenApiReference
            {
               Type = ReferenceType.SecurityScheme,
               Id = "Bearer"
            },
            Scheme = "oauth2",
            Name = "Bearer",
            In = ParameterLocation.Header,
         },
         new List<string>()
      }
   });
});
```

```
var key = Encoding.ASCII.GetBytes("mysuperdupersecret");
builder.Services.AddAuthentication(JwtBearerDefaults.AuthenticationScheme)
   .AddJwtBearer(options =>
   {
      options.TokenValidationParameters = new TokenValidationParameters
      {
         ValidateIssuer = true,
         ValidateAudience = true,
         ValidateLifetime = true,
         ValidateIssuerSigningKey = true,
         ValidIssuer = "mySystem",
         ValidAudience = "myUsers",
         IssuerSigningKey = new SymmetricSecurityKey(key)
      };
        });

builder.Services.AddScoped<CustomAuthFilter>();

var app = builder.Build();

app.UseSwagger();
app.UseSwaggerUI();

app.UseHttpsRedirection();

app.UseAuthentication();
app.UseAuthorization();

app.MapControllers();

app.Run();
```

- **Output**:



Exercise-3: Create Model & Custom Filter

- **Code**:

//Models/Employee.cs

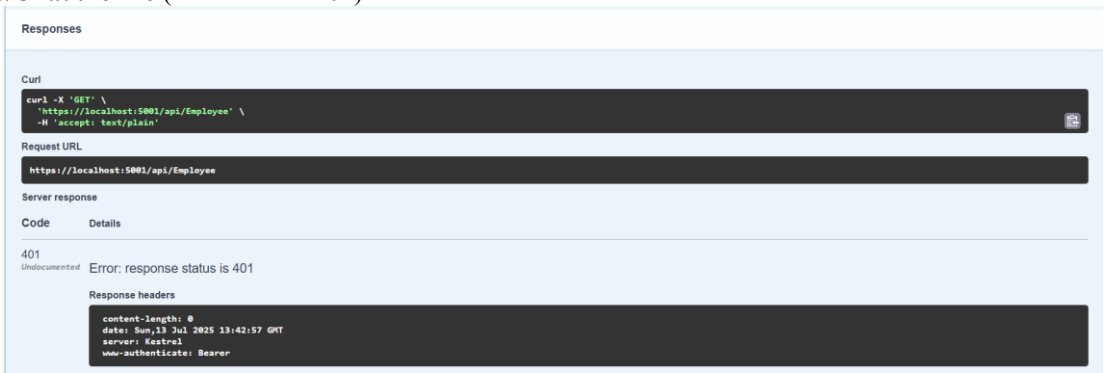
```
using System;
using System.Collections.Generic;

namespace WebApiDemo.Models
{
   public class Skill { public string SkillName { get; set; } }
      public class Department { public string DeptName { get; set; } }

   public class Employee
   {
      public int Id { get; set; }
      public string Name { get; set; }
      public int Salary { get; set; }
      public bool Permanent { get; set; }
      public Department Department { get; set; }
      public List<Skill> Skills { get; set; }
      public DateTime DateOfBirth { get; set; }
   }
}
```

//Filters/CustomAuthFilter.cs

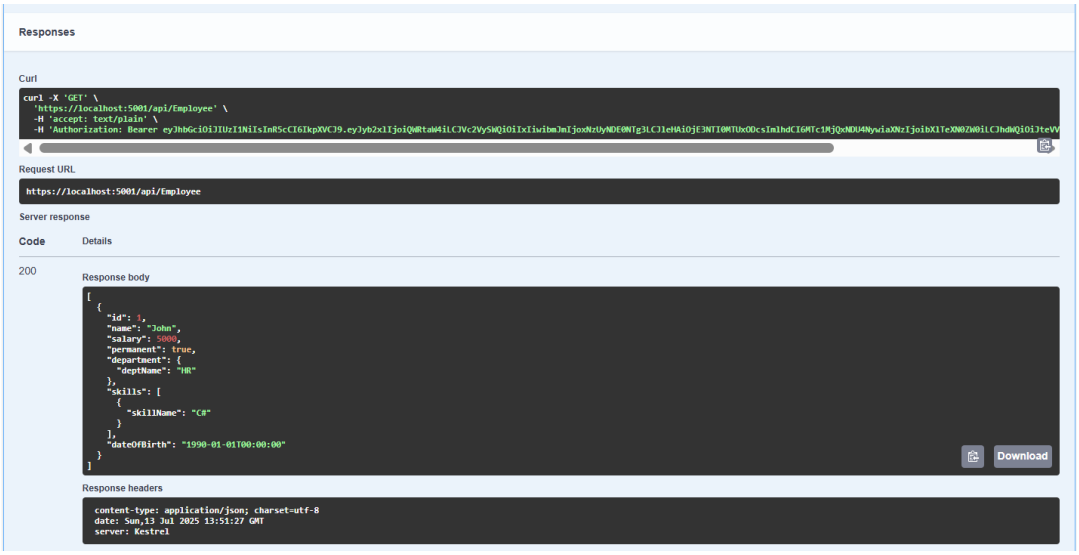
```
using Microsoft.AspNetCore.Mvc;
using Microsoft.AspNetCore.Mvc.Filters;

namespace WebApiDemo.Filters
{
   public class CustomAuthFilter : ActionFilterAttribute
   {
      public override void OnActionExecuting(ActionExecutingContext context)
      {
         if (!context.HttpContext.Request.Headers.TryGetValue("Authorization", out var token) ||
            !token.ToString().Contains("Bearer"))
         {
            context.Result = new BadRequestObjectResult("Invalid request - No Auth token or Bearer missing");
         }
      }
   }
}
```

- **<u>Output</u>:**

//**Unauthorize** (Status Code: 401)



//**Authorize** (Status Code: 200)



Exercise-4: Add PUT & DELETE for Employee API

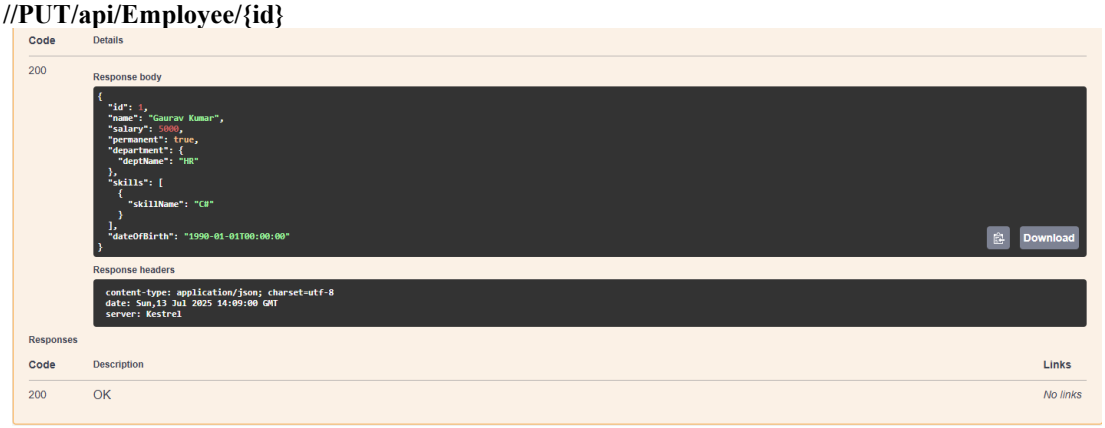
- **Code:**

```
using Microsoft.AspNetCore.Mvc;
using Microsoft.AspNetCore.Authorization;
using WebApiDemo.Models;
using System.Collections.Generic;
using System.Linq;
using System;

namespace WebApiDemo.Controllers
{
  [ApiController]
  [Route("api/[controller]")]
  [Authorize(Roles = "Admin")]
  public class EmployeeController : ControllerBase
  {
    private static List<Employee> employees = new()
    {
      new Employee
      {
        Id = 1,
        Name = "John",
        Salary = 5000,
        Permanent = true,
        Department = new Department { DeptName = "HR" },
        Skills = new List<Skill> { new Skill { SkillName = "C#" } },
        DateOfBirth = new DateTime(1990, 1, 1)
      }
    };

    [HttpGet]
    public ActionResult<List<Employee>> Get() => employees;

    [HttpPost]
    public IActionResult Post([FromBody] Employee emp)
    {
      employees.Add(emp);
      return Ok(emp);
    }

    [HttpPut("{id}")]
    public IActionResult Put(int id, [FromBody] Employee emp)
    {
      if (id <= 0)
```

```
            return BadRequest("Invalid employee id");

        var existing = employees.FirstOrDefault(e => e.Id == id);
        if (existing == null)
            return BadRequest("Invalid employee id");

        existing.Name = emp.Name;
        return Ok(existing);
    }

    [HttpDelete("{id}")]
    public IActionResult Delete(int id)
    {
        var existing = employees.FirstOrDefault(e => e.Id == id);
        if (existing == null)
            return NotFound();

        employees.Remove(existing);
        return NoContent();
    }
  }
}
```

- **Output:**

//PUT/api/Employee/{id}



//DELETE/api/Employee/{id}

Exercise-5: Implement JWT Token Authentication

- **Code:**

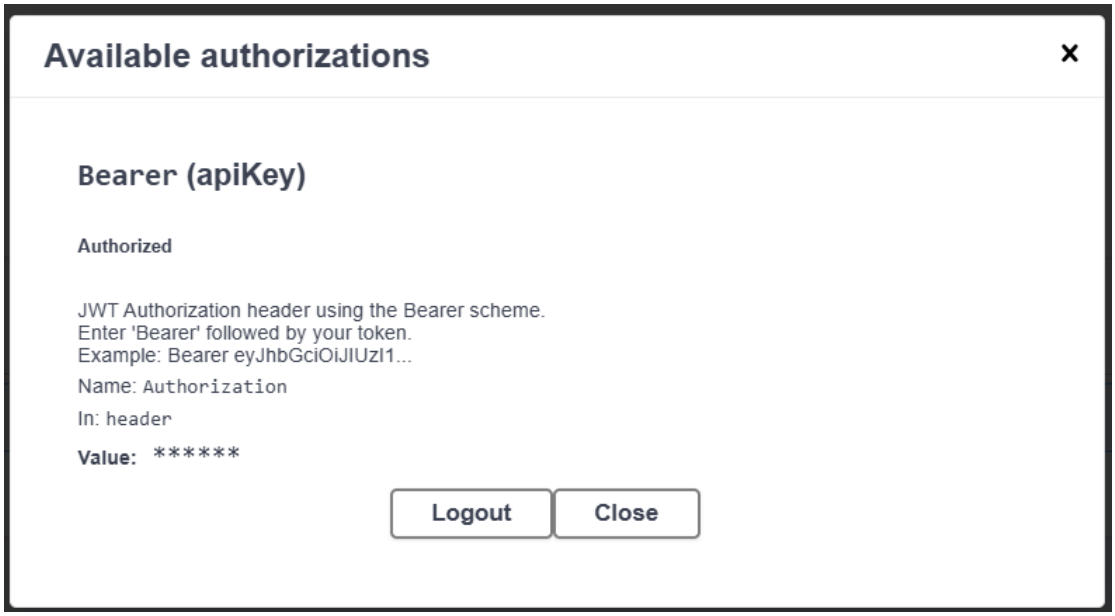
```
using Microsoft.AspNetCore.Mvc;
using Microsoft.AspNetCore.Authorization;
```

```
using Microsoft.IdentityModel.Tokens;
using System.Text;
using System.IdentityModel.Tokens.Jwt;
using System.Security.Claims;
using System;

namespace WebApiDemo.Controllers
{
  [ApiController]
  [Route("api/[controller]")]
  [AllowAnonymous]
  public class AuthController : ControllerBase
  {
    [HttpGet]
    public string GetToken()
    {
      var key = Encoding.ASCII.GetBytes("mysuperdupersecret");
      var tokenHandler = new JwtSecurityTokenHandler();

      var tokenDescriptor = new SecurityTokenDescriptor
      {
        Subject = new ClaimsIdentity(new[]
        {
          new Claim(ClaimTypes.Role, "Admin"),
          new Claim("UserId", "1")
        }),
        Expires = DateTime.UtcNow.AddMinutes(10),
        Issuer = "mySystem",
        Audience = "myUsers",
        SigningCredentials = new SigningCredentials(new SymmetricSecurityKey(key),
SecurityAlgorithms.HmacSha256Signature)
      };

      var token = tokenHandler.CreateToken(tokenDescriptor);
      return tokenHandler.WriteToken(token);
    }
  }
}
```

- **Output:**

//Token Generation

//Authorization



Exercise-6: Kafka Producer and Consumer

- **Code**:

//Kafka/KafkaProducer.cs

```
using Confluent.Kafka;
using System;
using System.Threading.Tasks;

namespace WebApiDemo.Kafka
{
  public class KafkaProducer
  {
    public static async Task Produce(string topic, string message)
    {
      var config = new ProducerConfig { BootstrapServers = "localhost:9092" };

      using var producer = new ProducerBuilder<Null, string>(config).Build();
      var result = await producer.ProduceAsync(topic, new Message<Null, string> { Value = message });
      Console.WriteLine($"Delivered to: {result.TopicPartitionOffset}");
    }
  }
}
```

//Kafka/KafkaConsumers.cs

```
using Confluent.Kafka;
using System;
using System.Threading;

namespace WebApiDemo.Kafka
{
  public class KafkaConsumer
  {
    public static void Consume(string topic)
    {
      var config = new ConsumerConfig
      {
        BootstrapServers = "localhost:9092",
        GroupId = "test-consumer-group",
        AutoOffsetReset = AutoOffsetReset.Earliest
      };
```

```
        using var consumer = new ConsumerBuilder<Ignore, string>(config).Build();
        consumer.Subscribe(topic);

        CancellationTokenSource cts = new CancellationTokenSource();

        try
        {
            while (true)
            {
                var cr = consumer.Consume(cts.Token);
                Console.WriteLine($"Consumed message: {cr.Value}");
            }
        }
        catch (OperationCanceledException)
        {
            consumer.Close();
        }
    }
  }
}
```

- **Output**:

//Producer

```
Delivered to: [my-topic][0]@offset 3
```

//Consumer

```
Consumed message: Hello from Swagger!
Consumed message: Another Kafka message
```